Supplementary material for

Coarse-Grained Models Using Local-Density Potentials Optimized with the Relative Entropy: Application to Implicit Solvation

Tanmoy Sanyal and M. Scott Shell

A. Relative sensitivity of the CG pair potential to inclusion of a local density field

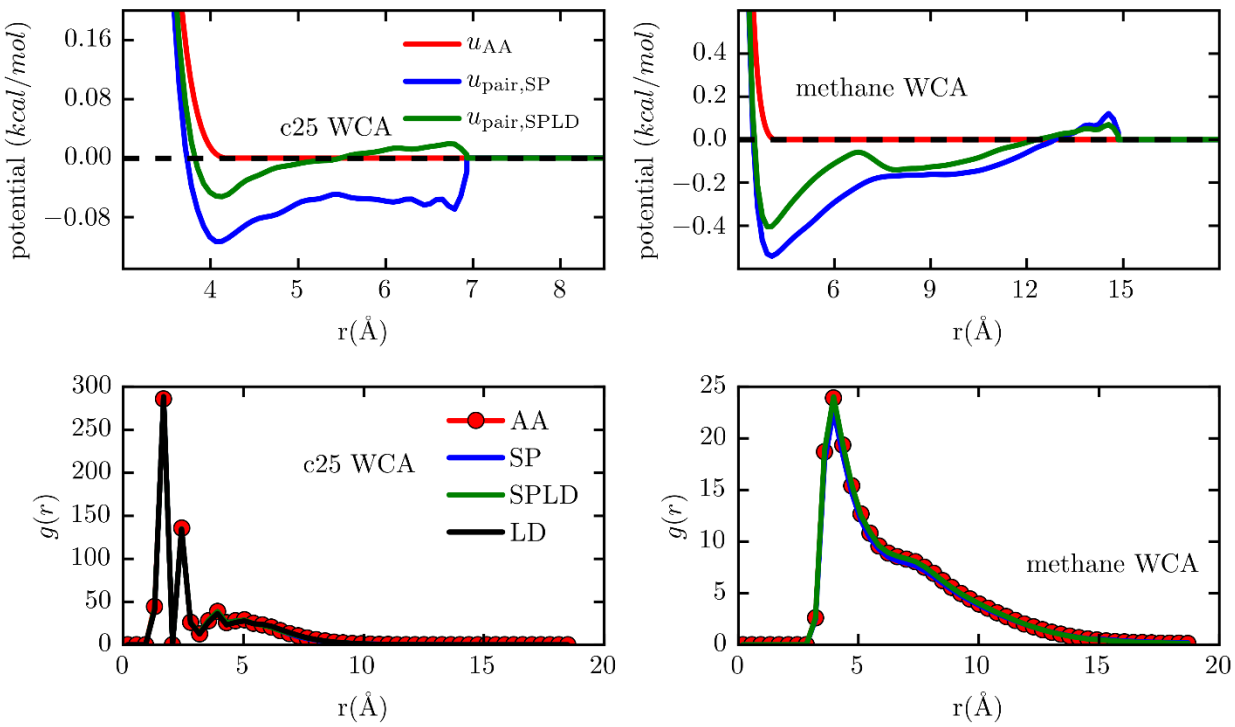

Figure S1 Upper panels show the pair part of the CG forcefield for the superhydrophobic polymer and free methane systems. For both, the inclusion of the LD potential (SPLD model) makes the CG pair potentials less attractive. Lower panels present a comparison of the intra-monomer radial distribution function between the AA system and the different CG models. For the polymer, even the LD-only model with no CG pair potential accurately reproduces the pair structure. (Note that all radial distributions approach zero because neither the monomers nor the methanes are bulk dispersed, the latter forming a dense cluster.)

Fig. S1 (upper panels) shows the sensitivity of the pair potential part of the CG forcefield for the superhydrophobic (WCA) polymer and methanes. The inclusion of the LD potential (SPLD model) makes the pair potential part of the forcefield less attractive, which is expected since the LD potential shares the (net) attractive manybody effect of the monomers that was previously sustained entirely by the pair potentials of the SP model (blue line). The lower panels demonstrate the intra-monomer pair correlation functions for the AA system and the different CG models. The SP and SPLD models capture these correlations equally well for both the polymer and the methane. For the polymer, even the LD model, in which pair potentials are not optimized, is able to reproduce the radial distribution function quantitatively. Note that the LD-only model was not optimized for the free methanes, based on our observation that the LD potential functions best only when supported by CG pair potentials.

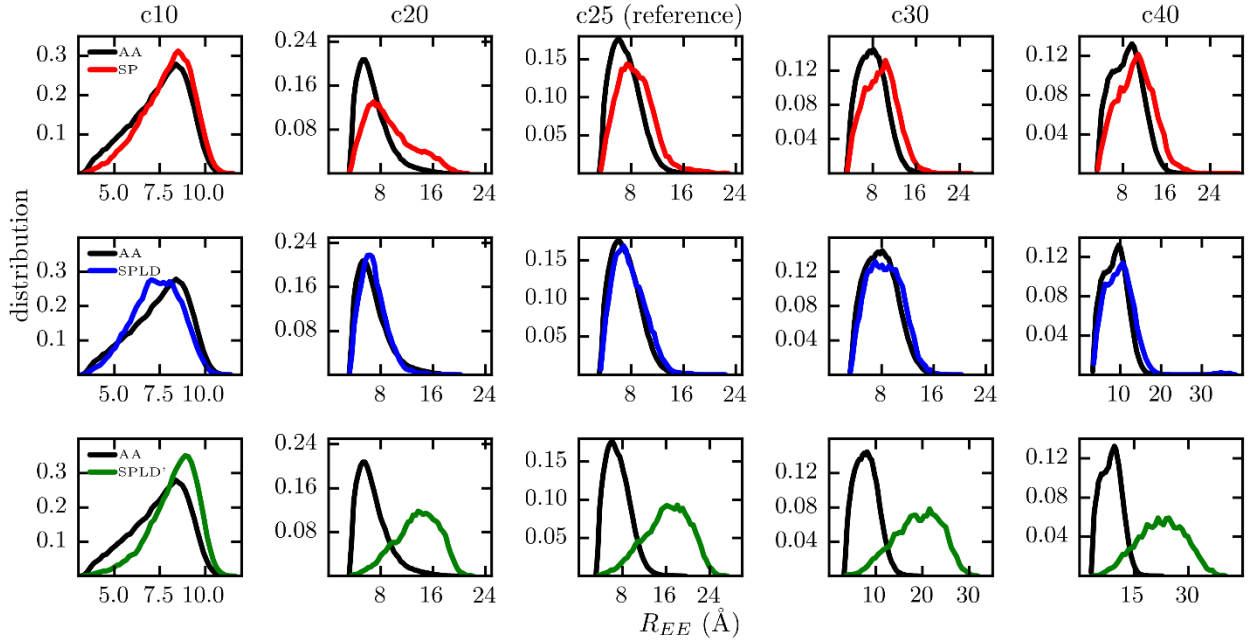

Figure S2 Comparison of the relative transferabilities of the c25 superhydrophobic WCA polymer end-to-end distance for a special forcefield (labelled SPLD') that considers only the CG pair component of the SPLD forcefield, with that for the SP and the SPLD models. Clearly, the pair component, although different from the SP model, produces poor transferability by mispredicting the distribution peaks.

Fig. S2 compares the R_{EE} transferability of the SP and SPLD models to the pair component of the SPLD model (labelled SPLD'). Clearly, the pair component alone is incapable of producing good transferability for R_{EE}. Similar results are noted (not shown here) for Rg transferability with this forcefield. It may be noted that though the SP and the CG pair component of the SPLD models in Fig S1 (top left panel) are similar in shape (other than the difference in well depth) for low and moderate distances, they are remarkably different at distances near the cutoff. This may explain why the pair part of the SPLD model produces vastly different transferability from the SP model. In general, while the sensitivity of the renormalized pair potentials to the local density field may be straightforward to predict (typically), transferability of thermophysical properties derived from different CG models is usually highly nontrivial to anticipate.

Fig. S3 compares the pair potential part of the SP and SPLD CG models for the LJ polymer and the LJ free methanes. Due to the absence of significant manybody effects, the SP and SPLD pair potentials for the LJ polymer are nearly the same, i.e., the renormalized pair potential of the SP model (blue line) does not change much after incorporating the local density field. For the case of the free methanes, however, inclusion of the LD potential (green line) makes the renormalized pair part more attractive. This likely stems from the need to maintain a (net) repulsive inter-monomer potential similar to that in the AA system, in the absence of bonds (which provide a natural separation e.g., between the beads in the polymer). The LD potential for the free LJ methanes optimized through relative entropy minimization (not shown here), is largely repulsive at higher local densities and so the CG pair component compensates to control the overall repulsive behavior and keep it close to that of the original methane-water interaction in the AA system

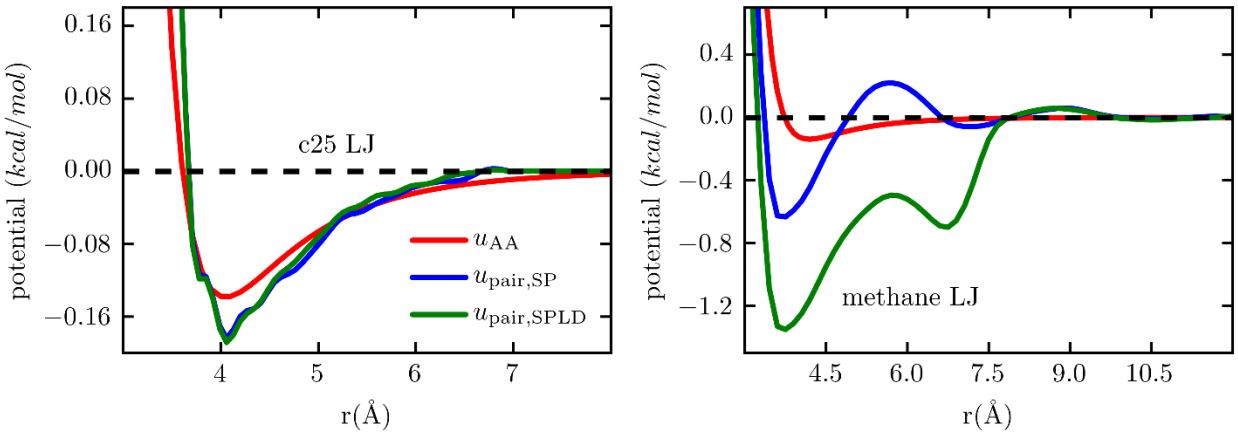

Figure S3 Sensitivity of the renormalized CG pair potential to the inclusion of the local density field for the LJ polymer (left) and free methanes (right). For the polymer, this sensitivity is low, presumably due to the lack of manybody effects. For the LJ methanes, however, relative entropy minimization produces a relatively repulsive LD potential (not shown here), which leads to more attractive character in the CG pair component of the SPLD model to maintain the same overall level of inter-monomer repulsion as that in the explicit water AA reference.

B. Local density distribution in C25 versus C40

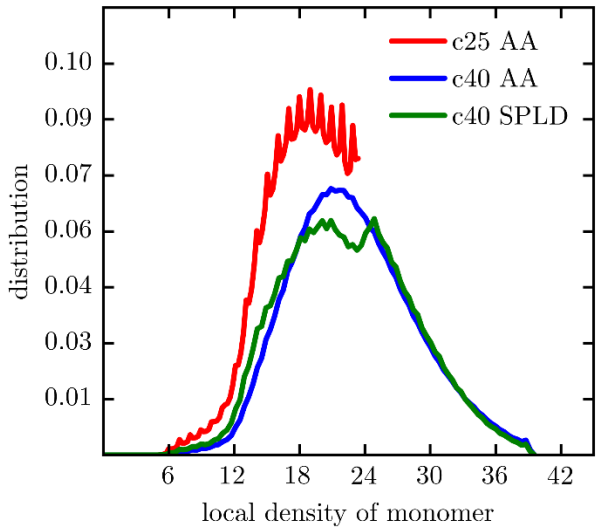

Figure S4 Comparison of the inter-monomeric local density distribution for the all atom trajectory of the superhydrophobic 25-mer with the all-atom and the SPLD models for a superhydrophobic 40-mer. The 25-mer embeds information about local density only up to 25 monomers which is significantly less than that conveyed by the 40-mer. The SPLD model is designed from the CG forcefield parameterized from the all atom 25-mer and thus fails to capture the peak of the true LD distribution for the 40-mer.

Figure S4 compares the distribution of local densities of the monomers for all atom c25 with the all atom and SPLD CG models for c40 (using the same LD cutoff, $r_c = 7.8$ Å). Although both the c25 and c40 LD distributions peak near 25 monomers, a significant portion of the distribution (~ 50 %) for c40 contains information about more than 25 monomers. Since the LD potential is multibody, the difference in spread (standard deviation) between the c25 and c40 distributions (in spite of possible agreement in the mean local densities) leads to non-trivial and difficult-to-intuit differences between their folding behavior. The SPLD model for c40, simulated using the forcefield parameterized from the all atom c25 trajectory lacks information beyond 25 monomers, which leads to inaccurate location of the distribution peak (green line).

C. Statistical uncertainty for the LD potential in the CG model of superhydrophobic methanes

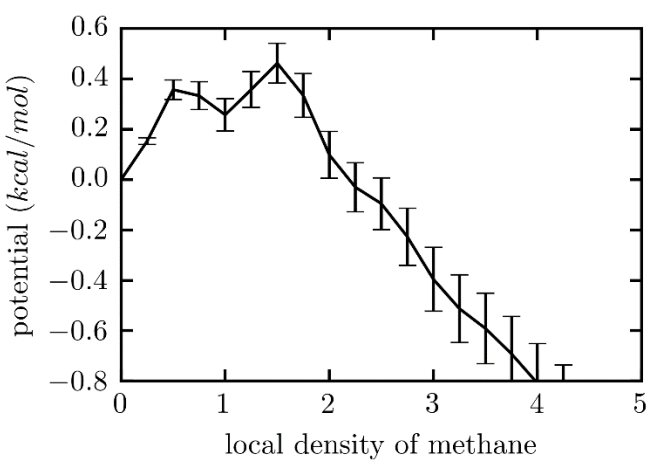

Figure S5 Statistical uncertainty of the local density potential in the SPLD model of the free methane system. The presence of the peaks at low local densities is not statistical artifact

The variation in the LD potential at low density in Fig. 11 is significant and does not reflect statistical uncertainty. To confirm, we re-optimized the LD potential for the SPLD model of the superhydrophobic methanes, starting from different initial values for the spline knots that

describe the LD potential. Fig. S5, presents the mean value of the LD potential from these runs, zoomed in on the low local density region. The two peaks are clearly outside the relative statistical uncertainty, which has an average value of 15 % (standard deviation normalized by the mean and averaged over the entire range of local densities). This has also been stated in the caption for Fig. 11 (page 31).

.